Jose J. Dominguez

Northglenn, CO | linkedin.com/in/jjosedominguez

Education

Colorado School of Mines Golden, CO
M.S. Mechanical Engineering Expected May 2027
B.S. Mechanical Engineering Expected May 2026

Work Experience and Research

Research Assistant - RoSE Lab Oct 2025 – Present
RoSE Lab - Space Exploration Department Golden, CO
Advisor: Bailey Hopkins `bailey_hopkins@mines.edu`

- Researching and building a functional prototype to measure net thrust force from rover wheels at different speeds and payload weight for Draw Bar Pull
- Maintained a rover to ensure reliable, high-performance operation
- Performed and recorded 1,600 rover mobility test simulations for Rover Mobility in the Mines Lunar Surface Simulator (MLSS)
- Prepared surfaces, ran experiments, and follow PPE guidelines for controlled environments culmination in an open source datasheet: Open Source Data Sheet - RoSE Lab

Front Desk Assistant Aug 2025 – Present
CSM – Computer Science Department Golden, CO
Supervisor: Justyna Glode `jglode@mines.edu`

- Served as first point of contact for students and staff; resolved routine requests and routed escalations
- Proofread emails and documents for clarity and correctness

Projects

CubeSAT – Senior Capstone Aug 2025 – Present
PI Advisor: Dr. Ian Jehn `ijehn@mines.edu`

- Creating a 10×10×11 cm, 2 kg CubeSat to detect space debris using RF and Wave-scattering signals from Star link satellites
- Power subsystem lead and CO-leading club members; ensuring solar panel and battery performance and standardized power regulation and distribution

Biomass Briquette (SolidWorks, Heat Transfer) Jan 2025 – May 2025

- Constructed a system that compresses and heats sawdust using a pneumatic pump to form cylindrical briquettes
- Coordinated with a team of six to research and refine an effective and efficient product

PID Easy-To-Bake Oven (LabVIEW, Heat Transfer) Aug 2024 – Dec 2024

- Developed a prototype using cardboard, aluminum, a thermocouple, and circuitry to successfully bake a cookie
- Implemented PID control with duty-cycle control and a mechanical relay; applied heat-transfer between materials to optimize temperature performance

Skills

Technologies and Professional: SolidWorks, Arduino IDE, MATLAB, Mathcad, LabVIEW, EES, Python, LaTeX, Ultimaker Cura, Microsoft Suite, FreeFlyer, C++, Bilingual(Spanish), Machine Shop Milling